315HW8

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GITHUB REPO

Problem 1

Part A

```
##       1
## 113.723
```

We use the fitted linear regression model:

creatclear =b0+b1 x age

Using the estimated coefficients from the model, we plug in age = 55:

Part B

```
##        age
## -0.6198159
```

The slope of the regression line (coefficient on age) represents the change in clearance per year of age. This tells us that creatinine clearance decreases by approximately 0.62 mL/min per year of age.

Part C

```
## [1] "40-year-old is healthier"
```

We predict expected values for both ages using the model and compute residuals (actual - predicted). The person with the larger positive residual is healthier relative to their expected value:

if (res_40 > res_60) "40-year-old is healthier" else "60-year-old is healthier"

This statement prints the correctly modeled statement

Problem 2

Part 1 - Italy

```
## $growth_rate
## days_since_first_death
##                  0.183
##
## $growth_ci
##  2.5% 97.5%
## 0.156 0.210
##
## $doubling_time
## days_since_first_death
##                    3.8
##
## $doubling_ci
##  2.5% 97.5%
##   4.4   3.3
```

We estimate the growth rate of COVID-19 deaths in Italy by fitting a linear model to the log-transformed daily deaths. We bootstrap by manually resampling the data 1000 times, refitting the model each time, and collecting the slope. The 95% confidence interval for the growth rate is taken from the 2.5th and 97.5th percentiles of the bootstrap distribution.

Part 2 - Spain

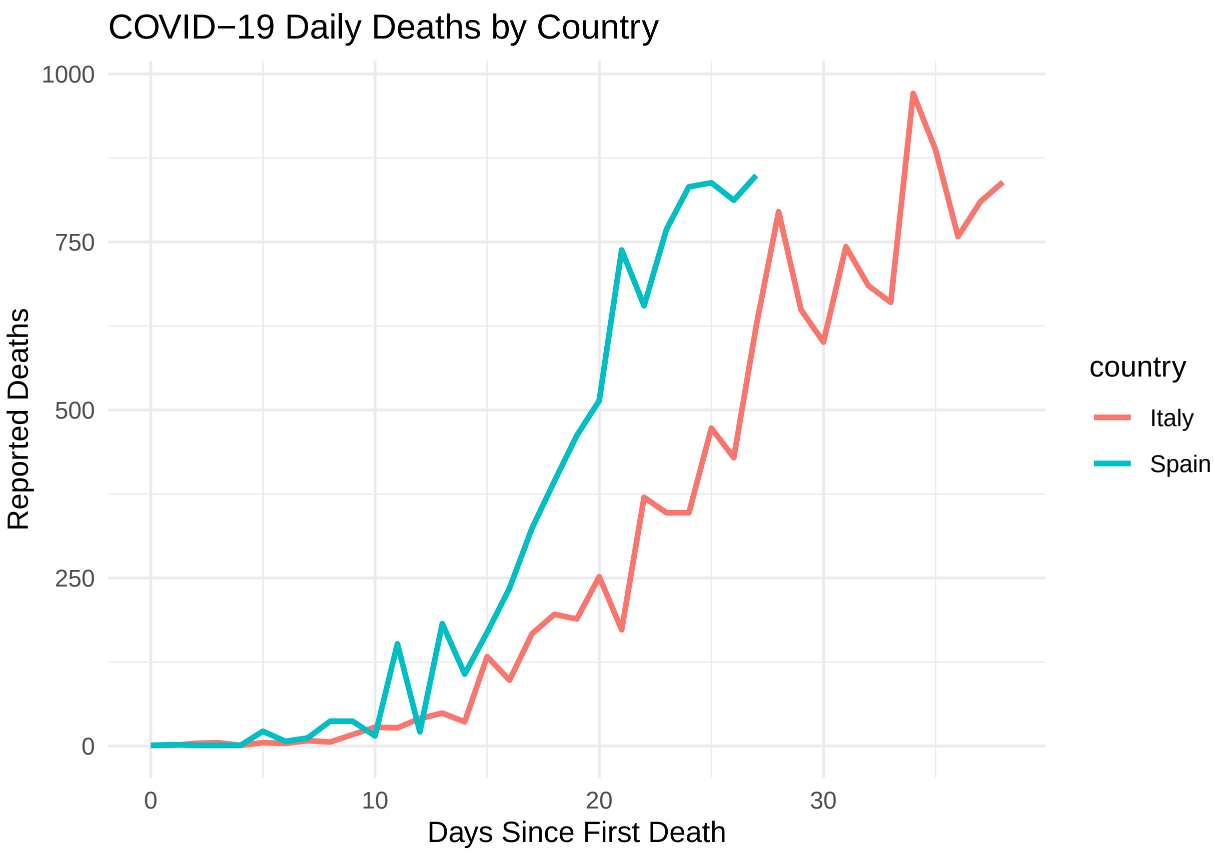
```
## $growth_rate
## days_since_first_death
##                  0.276
##
## $growth_ci
##  2.5% 97.5%
## 0.237 0.316
##
## $doubling_time
## days_since_first_death
##                    2.5
##
## $doubling_ci
##  2.5% 97.5%
##   2.9   2.2
```

The same bootstrapping process is used for Spain. Each iteration involves resampling the dataset with replacement, refitting the model, and storing the slope. This gives us a distribution of growth rate estimates from which we compute confidence intervals.

Part 3

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
```

```
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Problem 3

```
## $elasticity
## price_elasticity
##           -1.619
##
## $ci
##   2.5%  97.5%
## -1.775 -1.465
```

To estimate the price elasticity of demand for milk, I fit a model that relates the quantity of milk purchased to its price using a power-law relationship:

Q = KP^b, where b is the elasticity

by fitting this model to the data. To measure uncertainty, I performed bootstrapping by repeatedly sampling the data with replacement 1000 times and refitting the model each time to get a distribution of elasticity estimates.